

# Multi-Source Candidate Data Transformer

## Technical Design (One Page)

|                   |                                                                                                                   |
|-------------------|-------------------------------------------------------------------------------------------------------------------|
| Problem Statement | Transform structured (CSV/ATS) and unstructured (LinkedIn, GitHub, Resume, Notes) candidate data                  |
| Architecture      | Inputs → Extraction → Normalization → Merge → Conflict Resolution → Confidence → Provenance → Export              |
| Pipeline          | Extract → Normalize → Deduplicate → Merge → Confidence → Provenance → Validate → Export                           |
| Canonical Schema  | candidate_id, full_name, emails, phones, location, links, headline, experience, education, skills, provenance     |
| Merge Strategy    | Merge unique records, remove duplicates, prioritize trusted sources, preserve provenance.                         |
| Confidence        | Recruiter CSV(0.95) > ATS(0.90) > Resume(0.85) > LinkedIn(0.80) > GitHub(0.75) > Notes(0.60).                     |
| Provenance        | Each output field stores source, confidence score and merge method for complete traceability.                     |
| Runtime Config    | Configurable fields, field renaming, normalization toggle, confidence inclusion and missing-value handling        |
| Edge Cases        | Missing files, malformed JSON/CSV, duplicate values, inconsistent dates, phone formats, skill spelling variations |
| Assumptions       | All inputs refer to one candidate, UTF-8 encoded, predefined source priorities, minimum one structured source     |